Resilience Under Pressure: A Data-Driven Framework for Global Disaster Risk, Response, and Recovery Analytics

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***Abstract*—Natural and man-made disasters have increasingly exposed vulnerabilities in governance, economic systems, and infrastructure. This report introduces a data-driven framework for analyzing disaster resilience through computational models and visual analytics. Derived metrics such as Disaster Impact Index (DII), Resilience Recovery Score (RRS), and Composite Resilience Index (CRI) are implemented to quantify resilience across nations and regions. By integrating datasets like EM-DAT, World Bank Indicators, and HDR data, the analysis provides actionable insights for policymakers. The framework emphasizes the importance of socio-economic and institutional factors in disaster recovery and resilience, offering a comprehensive approach to understanding and mitigating disaster impacts. Key findings highlight the role of governance and adaptive capacity in shaping resilience outcomes, with visualizations enabling exploration of trends across time, geography, and disaster types.**

I. Problem Statement and Objectives

Disaster resilience is multi-dimensional and cannot be directly measured. This project aims to:

- Define computational metrics for resilience.
- Integrate datasets across socio-economic and environmental domains.
- Develop visual analytics to explore resilience trends and drivers.

II. Data Sources and Preprocessing

A. Datasets Used

- **EM-DAT Disaster Database:** https://www.emdat.be/
- **World Bank Indicators:** https://data.worldbank.org/
- **HDR Data:** http://hdr.undp.org/en/data
- **Our World in Data:** https://ourworldindata.org/natural-disasters
- **World Governance Indicators (WGI):** https://www.worldbank.org/en/publication/worldwide-governance-indicators
- **Urban Population Data:** https://data.worldbank.org/indicator/SP.URB.TOTL.IN.ZS

B. Preprocessing Steps

- **Standardizing Country Names:** Applied a mapping to ensure consistent country names across datasets.
- **Handling Missing Values:** Addressed missing values analytically through imputation or removal.
- **Renaming Columns:** Standardized column names to align attributes across datasets.
- **Data Type Conversion:** Converted columns to appropriate data types (e.g., numeric, categorical).
- **Merging Datasets:** Combined six datasets (EM-DAT, World Bank Indicators, HDR Data, OWID, WGI, Urban Population) on country-year pairs using left joins.
- **Dropping Invalid Rows:** Removed rows with missing country-year pairs to ensure data integrity.

III. Methods and Implementation

A. Feature Engineering

Derived attributes include:

- **Severity Weight (S):** Based on disaster type.
- **Human Cost Ratio:** Fatalities per 100k population.
- **Affected Population Percentage:** Percentage of population affected.
- **Economic Loss Normalized by GDP:** Economic loss as a percentage of GDP.
- **Annualized Disaster Frequency:** Number of disasters per country-year.
- **GDP Growth Rate Change:** Recovery rate proxy (post - pre growth).
- **Infrastructure Exposure:** Urbanization × disaster frequency.
- **Disaster Impact Index (DII):** Composite metric for disaster impact.
- **Resilience Recovery Score (RRS):** Simplified resilience recovery metric.
- **Adaptive Capacity (A):** Composite of HDI, governance, and urban infrastructure.
- **Exposure (E):** Disaster frequency × average severity.
- **Vulnerability (V):** Normalized disaster impact index.
- **Composite Resilience Index (CRI):** Adaptive capacity divided by exposure × vulnerability.
- **Data Quality Flags:** Indicators for missing data in key attributes.

B. Computational Models

- **Disaster Impact Index (DII):**

$$\text{DII} = \frac{(F + Apop)}{GDPpc} \times S$$

- **Resilience Recovery Score (RRS):**

$$\text{RRS} = \frac{GDPgrowthpost - GDPgrowthpre}{Trecovery} + \frac{HDI + Govindex}{2}$$

- **Composite Resilience Index (CRI):**

$$\text{CRI} = \frac{A}{E \times V}$$

C. Algorithms, Visualization Tools, and Libraries

In this project, we did not use any machine learning algorithms or external libraries for analysis. Instead, we relied on Tableau as our primary visualization tool. Tableau was chosen for its ability to create interactive and dynamic dashboards, which are ideal for exploring complex datasets like those used in this project. The platform allowed us to design visualizations that effectively communicate insights, such as trends in resilience, exposure, and socio-economic factors. Tableau's drag-and-drop interface enabled rapid prototyping and iteration, while its interactivity empowered users to explore data from multiple perspectives. This made Tableau the best choice for visualizing the relationships between disaster resilience metrics and other variables.

IV. Key Results, Visualizations, and Interpretation

A. Dashboard Overview

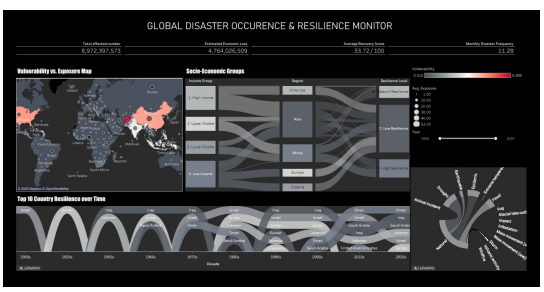


Fig. 1. Interactive Dashboard Overview

The interactive dashboard provides a comprehensive view of disaster resilience metrics across countries and regions. It includes multiple analytical views, such as geo-maps, time-series plots, and scatter plots, enabling users to explore resilience trends over time and geography. Filters allow users to focus on specific disaster types, socio-economic groups, or time periods. For example, the dashboard highlights countries with high exposure but low resilience, emphasizing the need for targeted interventions. It also shows how governance and socio-economic factors influence recovery rates, with wealthier nations often recovering faster. The interactivity of the dashboard allows policymakers to identify key resilience drivers and design data-driven strategies to reduce disaster risk.

B. Vulnerability vs Exposure Map

Fig. 2. Vulnerability vs Exposure Map

This map highlights the relationship between resilience and exposure across different countries. Countries with high exposure but low resilience are at greater risk, while those with high resilience can better withstand disasters. The visualization emphasizes the need for targeted interventions in regions with low resilience.

C. Top 10 Country Resilience Over Time (Bump Chart)

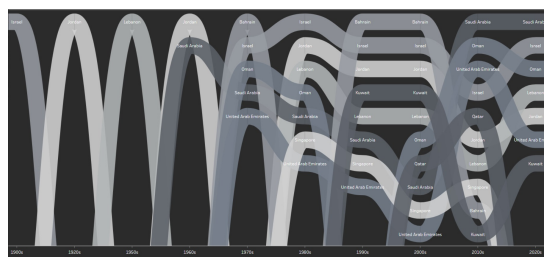


Fig. 3. Top 10 Country Resilience Over Time

This bump chart shows the resilience rankings of the top 10 countries over time. It reveals trends in resilience improvement or decline, highlighting the effectiveness of governance and socio-economic policies in shaping resilience outcomes.

D. Disaster Types Flow (Radial Sankey Chart)

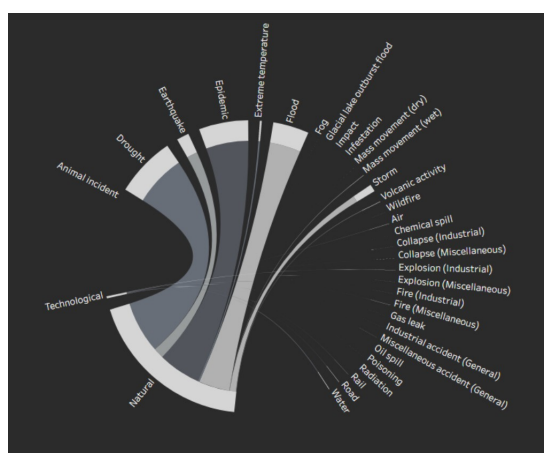


Fig. 4. Disaster Types Flow (Radial Sankey Chart)

The radial Sankey chart illustrates the flow of different disaster types and their impacts. It provides a clear view of which disaster types contribute most to overall exposure and resilience challenges, enabling policymakers to prioritize resources effectively.

E. Socio-Economic Groups (Sankey Graph)

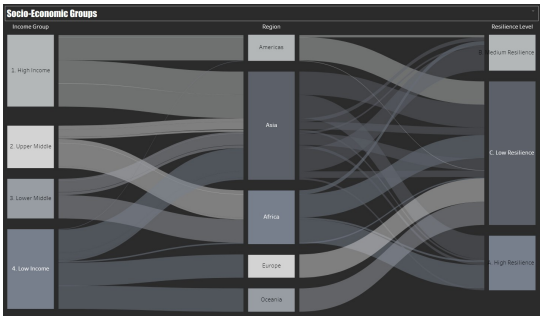


Fig. 5. Socio-Economic Groups (Sankey Graph)

This Sankey graph explores the connections between socio-economic groups and resilience metrics. It highlights disparities in resilience across different socio-economic categories, emphasizing the role of equity in disaster recovery and preparedness.

F. Insights and Findings

The visualizations collectively provide several key insights:

- Countries with high resilience recover faster from disasters, even when exposure is significant.
- Governance and socio-economic factors play a critical role in shaping resilience outcomes.
- Certain disaster types, such as floods and earthquakes, have a disproportionate impact on resilience metrics.
- Socio-economic disparities significantly influence resilience, underscoring the need for equitable policies.

These findings offer actionable insights for policymakers, enabling them to design targeted interventions to enhance resilience and reduce disaster risks.

V. Limitations And Future Work

- The free version of Tableau lacks advanced features, limiting the scope of visualizations.
- Data availability was inconsistent across regions, leading to potential biases in analysis.
- Some datasets had missing values, which required imputation and may have affected accuracy.

- Focus more on Middle Eastern countries as they have the highest resilience.
- Incorporate additional datasets, such as climate change indicators, to enhance the analysis.
- Develop predictive models to forecast resilience trends based on socio-economic and governance factors.

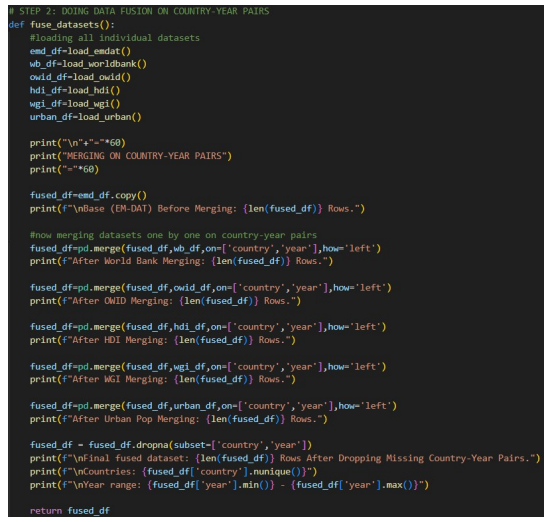
VI. Appendix

A. Additional Plots and Tables

TABLE I
Sample Data from Merged Dataset

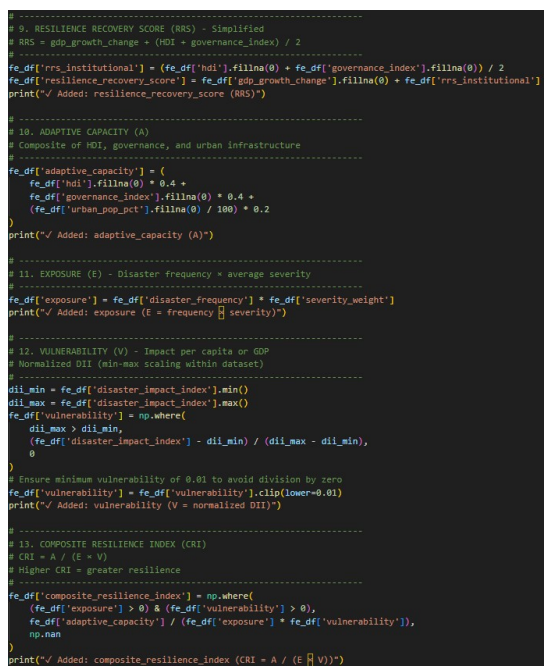
Country	Year	Fatalities	Affected	Economic Loss
Afghanistan	1954	2000	0	0
Afghanistan	1956	100	2000	25000
Afghanistan	1963	107	0	0
Afghanistan	1964	74	400	400

B. Code Snippet Screenshots



```python
# STEP 2: DOING DATA FUSION ON COUNTRY-YEAR PAIRS
def fuse_datasets():
    #loading all individual datasets
    emd_df=load_emdat()
    wb_df=load_worldbank()
    owid_df=load_owid()
    hdi_df=load_hdi()
    wgi_df=load_wgi()
    urban_df=load_urban()

    print("\n"+"="*60)
    print("MERGING ON COUNTRY-YEAR PAIRS")
    print("="*60)

    fused_df=emd_df.copy()
    print(f"\nBase (EM-DAT) Before Merging: {len(fused_df)} Rows.")

    #now merging datasets one by one on country-year pairs
    fused_df=pd.merge(fused_df,wb_df,on=['country','year'],how='left')
    print(f"After World Bank Merging: {len(fused_df)} Rows.")

    fused_df=pd.merge(fused_df,owid_df,on=['country','year'],how='left')
    print(f"After OWID Merging: {len(fused_df)} Rows.")

    fused_df=pd.merge(fused_df,hdi_df,on=['country','year'],how='left')
    print(f"After HDI Merging: {len(fused_df)} Rows.")

    fused_df=pd.merge(fused_df,wgi_df,on=['country','year'],how='left')
    print(f"After WGI Merging: {len(fused_df)} Rows.")

    fused_df=pd.merge(fused_df,urban_df,on=['country','year'],how='left')
    print(f"After Urban Pop Merging: {len(fused_df)} Rows.")

    fused_df = fused_df.dropna(subset=['country','year'])
    print(f"\nFinal fused dataset: {len(fused_df)} Rows After Dropping Missing Country-Year Pairs.")
    print(f"\nCountries: {fused_df['country'].nunique()}")
    print(f"\nYear range: {fused_df['year'].min()} - {fused_df['year'].max()}")

    return fused_df
```

Fig. 6. Screenshot - Merging Datasets



```python
# ----------------------------------------------------------------
# 9. RESILIENCE RECOVERY SCORE (RRS) - Simplified
# RRS = gdp_growth_change + (HDI + governance_index) / 2
# ----------------------------------------------------------------
fe_df['rrs_institutional'] = (fe_df['hdi'].fillna(0) + fe_df['governance_index'].fillna(0)) / 2
fe_df['resilience_recovery_score'] = fe_df['gdp_growth_change'].fillna(0) + fe_df['rrs_institutional']
print("✓ Added: resilience_recovery_score (RRS)")

# ----------------------------------------------------------------
# 10. ADAPTIVE CAPACITY (A)
# Composite of HDI, governance, and urban infrastructure
# ----------------------------------------------------------------
fe_df['adaptive_capacity'] = (
    fe_df['hdi'].fillna(0) * 0.4 +
    fe_df['governance_index'].fillna(0) * 0.4 +
    (fe_df['urban_pop_pct'].fillna(0) / 100) * 0.2
)
print("✓ Added: adaptive_capacity (A)")

# ----------------------------------------------------------------
# 11. EXPOSURE (E) - Disaster frequency × average severity
# ----------------------------------------------------------------
fe_df['exposure'] = fe_df['disaster_frequency'] * fe_df['severity_weight']
print("✓ Added: exposure (E = frequency × severity)")

# ----------------------------------------------------------------
# 12. VULNERABILITY (V) - Impact per capita or GDP
# Normalized DII (min-max scaling within dataset)
# ----------------------------------------------------------------
dii_min = fe_df['disaster_impact_index'].min()
dii_max = fe_df['disaster_impact_index'].max()
fe_df['vulnerability'] = np.where(
    dii_max > dii_min,
    (fe_df['disaster_impact_index'] - dii_min) / (dii_max - dii_min),
    0
)
# Ensure minimum vulnerability of 0.01 to avoid division by zero
fe_df['vulnerability'] = fe_df['vulnerability'].clip(lower=0.01)
print("✓ Added: vulnerability (V = normalized DII)")

# ----------------------------------------------------------------
# 13. COMPOSITE RESILIENCE INDEX (CRI)
# CRI = A / (E × V)
# Higher CRI = greater resilience
# ----------------------------------------------------------------
fe_df['composite_resilience_index'] = np.where(
    (fe_df['exposure'] > 0) & (fe_df['vulnerability'] > 0),
    fe_df['adaptive_capacity'] / (fe_df['exposure'] * fe_df['vulnerability']),
    np.nan
)
print("✓ Added: composite_resilience_index (CRI = A / (E × V))")
```

Fig. 7. Screenshot - Feature Engineering [Important Metrics]

```python
# main func
if __name__ == "__main__":
    print("="*60)
    print("RESILIENCE UNDER PRESSURE DATA FUSION PROCESSING")
    print("="*60)

    # Step 1-2: Load and fuse datasets
    fused_df = fuse_datasets()

    # Step 3: Feature engineering
    final_df = engineer_features(fused_df)

    # Step 4: Save to CSV
    save_to_csv(final_df)

    # Print summaries
    print_feature_summary(final_df)

    print("\n"+"="*60)
    print("SAMPLE DATA (First 5 rows):")
    print("="*60)
    print(final_df.head())

    print("\n"+"="*60)
    print("ALL COLUMNS IN FINAL DATASET:")
    print("="*60)
    print(final_df.columns.tolist())

    print("\n"+"="*60)
    print("NEW ENGINEERED FEATURES:")
    print("="*60)
    new_features = [
        'severity_weight', 'human_cost_ratio', 'affected_pop_pct',
        'economic_loss_gdp_pct', 'disaster_frequency', 'gdp_growth_change',
        'infrastructure_exposure', 'fatalities_per_million', 'disaster_impact_index',
        'resilience_recovery_score', 'adaptive_capacity', 'exposure',
        'vulnerability', 'composite_resilience_index', 'data_quality_score'
    ]
    for f in new_features:
        print(f"  • {f}")

    print("\n"+"="*60)
    print("FUSION PROCESS COMPLETE!")
    print("="*60)
    print(f"\nFinal dataset saved to: {OUTPUT_FILE}")
    print(f"Total rows: {len(final_df)}")
    print(f"Total columns: {len(final_df.columns)}")
```

Fig. 8. Screenshot - Flow of the Project Implementation

VII. References

- EM-DAT Disaster Database: https://www.emdat.be/
- World Bank Indicators: https://data.worldbank.org/
- HDR Data: http://hdr.undp.org/en/data
- Our World in Data: https://ourworldindata.org/natural-disasters
- World Governance Indicators (WGI): https://www.worldbank.org/en/publication/worldwide-governance-indicators
- Urban Population Data :https://data.worldbank.org/indicator/SP.URB.TOTL.IN.ZS